

CHE 201 Spring 2020 HW 1

Anna Christensen

TOTAL POINTS

96 / 100

QUESTION 1

1 Question 1 10 / 10

✓ - **0 pts** Correct

QUESTION 2

2 Question 2 20 / 20

✓ - **0 pts** Correct

- **3 pts** Wrong occurrences within the dishwasher system.

- **3 pts** Wrong occurrences within the shower system.

- **4 pts** Description could be more detailed.

- **3 pts** Missing an identification of the location of a system boundary.

- **3 pts** Wrong occurrences within the water heater system.

- **9 pts** Missing identifications of the locations of the system boundaries.

QUESTION 3

3 Question 3 10 / 10

- **3 pts** Answers of b,d,h are wrong. Formulas are right. Just recalculate.

✓ - **0 pts** Corret

- **1 pts** Can not read a)

- **1 pts** Answer of e) is wrong. Formulas are right. Just recalculate.

- **1 pts** Answer of h) is wrong.

- **0.5 pts** Answer of d) need to be more precise.

QUESTION 4

4 Question 4 10 / 10

✓ - **0 pts** Correct

- **5 pts** Redo calculation of circled parts.

- **5 pts** Did not answer 4 b)

- **10 pts** No answer

QUESTION 5

5 Question 5 10 / 10

✓ - **0 pts** Correct

- **5 pts** Unit of B is wrong.

- **0 pts** You lost a "f" on A

- **5 pts** Unit of B is wrong. $[B]=[P]^*[V]$

- **2.5 pts** So what's the unit of B?

- **10 pts** No answer submitted.

QUESTION 6

6 Question 6 10 / 10

✓ - **0 pts** Correct

- **5 pts** Wrong judgement on vacuum and gage pressure.

- **5 pts** Wrong reading of Bourdon gage.

- **10 pts** No answer submitted.

QUESTION 7

7 Question 7 8 / 10

✓ - **2 pts** Wrong units of final results.

- **0 pts** correct

- **3 pts** Wrong judgement of pressure increase or decrease.

- **3 pts** Did not answer the second question, increase or decrease of pressure.

- **7 pts** Did not answer the pressure difference between a and b.

- **10 pts** No answer.

QUESTION 8

8 Question 8 18 / 20

- **0 pts** Well explained.

- **4 pts** Investigation on the physical basis of Korotkoff sound could be more detailed and specific.

- **4 pts** Investigation on the role of systolic and diastolic pressure on medical practice could be more detailed and specific.

- **2 pts** Investigation on the physical basis of Korotkoff sound could be more detailed and specific.

✓ - **2 pts** Investigation on the role of systolic and diastolic pressure on medical practice could be more detailed and specific.

- **2 pts** No references listed.

- **20 pts** No answer.

- **8 pts** No answer of the significance of these pressures on medical practice.

- **10 pts** The handwriting is hard to read. Also, the investigation could be more detailed and specific.

CHE 201 HW 1
Due: 11.59 pm on 01/25/2020

1. (10 points)

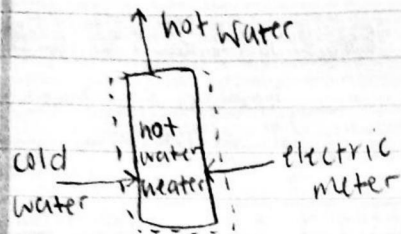
For problems 1.1–1.10, match the appropriate definition in the right column with each term in the left column.

- | | | |
|----------|---|--------------------|
| 1.1. WP | I | Boundary |
| 1.2. WP | G | Closed system |
| 1.3. WP | B | Control volume |
| 1.4. WP | E | Extensive property |
| 1.5. WP | H | Intensive property |
| 1.6. WP | D | Process |
| 1.7. WP | J | Property |
| 1.8. WP | A | State |
| 1.9. WP | F | Surroundings |
| 1.10. WP | C | System |
-
- | | |
|----|--|
| A. | The condition of a system as described by its properties |
| B. | A region of space through which mass may flow |
| C. | Whatever is studied |
| D. | A transformation from one state to another |
| E. | A property whose value for an overall system is the sum of its values for the parts into which the system is divided |
| F. | Everything external to the system |
| G. | A fixed quantity of matter |
| H. | A property whose value is independent of the size or extent of a system and may vary from place to place within the system at any moment |
| I. | Distinguishes the system from its surroundings |
| J. | A macroscopic characteristic of a system to which a numerical value can be assigned at a given time without knowledge of the previous behavior of the system |

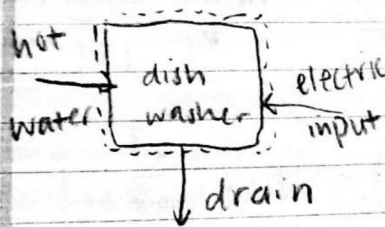
1 Question 1 10 / 10

✓ - 0 pts Correct

②

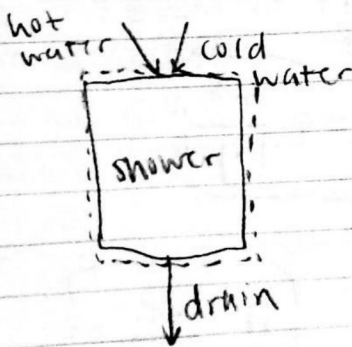


There are two points where mass flows into and out of the water heater system. Cold water enters the boundary and hot water exits the boundary. The electric meter inputs energy to heat the cold water.



There are also two points of mass flow for the dishwasher. Hot water enters the boundary, and the dirty water exits the boundary through the drain.

The electric inputs energy to run the dishwasher.



The hot and cold water enter the boundary of the shower system. The dirty water exits the system through the drain. There are 2 points of matter transfer.

2 Question 2 20 / 20

✓ - 0 pts Correct

- 3 pts Wrong occurrences within the dishwasher system.
- 3 pts Wrong occurrences within the shower system.
- 4 pts Description could be more detailed.
- 3 pts Missing an identification of the location of a system boundary.
- 3 pts Wrong occurrences within the water heater system.
- 9 pts Missing identifications of the locations of the system boundaries.

③ (a) 122 in^3 to L

$$\sqrt[3]{122 \text{ in}^3} = 4.96 \text{ in} = \frac{4.96 \text{ in}}{12 \text{ in/ft}} = 0.413 \text{ ft} = \cancel{19.07 \text{ ft}}$$

$$0.413 \text{ ft} \times \frac{0.3048 \text{ m}}{1 \text{ ft}} = (0.126 \text{ m})^3 = 0.00199 \text{ m}^3$$

$$0.00199 \text{ m}^3 \times \frac{1000 \text{ L}}{1 \text{ m}^3} = \boxed{1.99 \text{ L}}$$

(b) $778.17 \text{ ft} \cdot \text{lb}_f$ to kJ

$$778.17 \text{ ft} \cdot \text{lb}_f \times \frac{1 \text{ kJ}}{738 \text{ ft} \cdot \text{lb}_f} = \boxed{1.05 \text{ kJ}}$$

(c) 100 hp to kW

$$100 \text{ hp} \times \frac{1.0 \text{ kW}}{1.341 \text{ hp}} = \boxed{74.57 \text{ kW}}$$

(d) 1000 lb/h to kg/s

$$1000 \frac{\text{lb}}{\text{h}} \times \frac{0.45359 \text{ kg}}{1 \text{ lb}} \times \frac{1 \text{ hr}}{3600 \text{ s}} = \boxed{0.126 \frac{\text{kg}}{\text{s}}}$$

e) 29.392 lbf/in² to bar

$$29.392 \frac{\text{lbf}}{\text{in}^2} \times \frac{1.0 \frac{\text{lbf}}{\text{in}^2}}{14.504 \text{ bar}} = 2.03 \text{ bar}$$

f) 2500 $\frac{\text{ft}^3}{\text{min}}$ to $\frac{\text{m}^3}{\text{s}}$

$$2500 \frac{\text{ft}^3}{\text{min}} \times \frac{0.3048^3 \text{ m}^3}{\text{ft}^3} \times \frac{1 \text{ min}}{60 \text{ s}} = 1.18 \frac{\text{m}^3}{\text{s}}$$

g) 75 $\frac{\text{mile}}{\text{h}}$ to $\frac{\text{km}}{\text{h}}$

$$75 \frac{\text{mile}}{\text{h}} \times \frac{1.6 \text{ km}}{1 \text{ mile}} = 120 \frac{\text{km}}{\text{h}}$$

h) 1 ton to N
2000 lbf to N

$$2000 \text{ lbf} \times \frac{4.448 \text{ N}}{1 \text{ lbf}} = 8896 \text{ N}$$

3 Question 3 10 / 10

- **3 pts** Answers of b,d,h are wrong. Formulas are right. Just recalculate.

✓ - **0 pts** Corret

- **1 pts** Can not read a)

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- **1 pts** Answer of h) is wrong.

- **0.5 pts** Answer of d) need to be more precise.

④

0.3 kmol
octane

$$Vol = 5 m^3$$

find a) weight in N

b) molar and mass based
specific volumes $\frac{m^3}{kmol}$

$$\text{molar mass} = 114.23 \frac{g}{mol}$$

$$\text{a) } 114.23 \frac{g}{mol} \times \frac{1000 mol}{1 kmol} = 114230 \frac{g}{kmol} = 114.230 kg \frac{m^3}{kg}$$

$$0.3 kmol \times 114230 \frac{g}{kmol} = 34269 g \times \frac{1 kg}{1000 g} = 34.269 kg$$

$$F = ma = 34.269 kg (9.81 \frac{m}{s^2}) = \boxed{336.18 N}$$

⑥

$$\bar{V} = MV$$

$$\frac{m^3}{kmol} = \frac{kg}{kmol} \left(\frac{m^3}{kg} \right)$$

$$n = \frac{\text{mass (kg)}}{M (\frac{kg}{kmol})}$$

$$\bar{V} = \left(114.230 \frac{kg}{kmol} \right) \left(\frac{5 m^3}{34.269 kg} \right) = \boxed{16.67 \frac{m^3}{kmol}}$$

$$\rho = \frac{kg}{m^3} = \frac{34.269 kg}{5 m^3} = 6.85 \frac{kg}{m^3}$$

$$V = \frac{1}{\rho} = \frac{1}{6.8 \frac{kg}{m^3}} = \boxed{0.1459 \frac{m^3}{kg}}$$

4 Question 4 10 / 10

✓ - **0 pts** Correct

- **5 pts** Redo calculation of circled parts.

- **5 pts** Did not answer 4 b)

- **10 pts** No answer

$$⑤ \quad p = A + (B/V)$$

$$\frac{\text{lb}_f}{\text{ft}^2} = \frac{\text{lb}_f}{\text{ft}^2} + \frac{\text{lb}_f \cdot \text{ft}}{\text{ft}^{3/2}}$$

$$\begin{array}{|c|} \hline A \quad \frac{\text{lb}_f}{\text{ft}^2} \\ \hline B \quad \text{lb}_f \cdot \text{ft} \\ \hline \end{array}$$

$$⑥ \quad p_{\text{atm}} = 1 \text{ bar}$$

$$1 \text{ bar} \times \frac{100 \text{ kPa}}{1 \text{ bar}} = 100 \text{ kPa}$$

$$p_{\text{absolute}} = 200 \text{ kPa}$$

$$p_{\text{atm}} < p_{\text{absolute}}$$

$$p_{\text{gage}} = p_{\text{absolute}} - p_{\text{atm}}$$

$$p_{\text{gage}} = 200 \text{ kPa} - 100 \text{ kPa}$$

$$p_{\text{gage}} = 100 \text{ kPa}$$

This is a gage pressure because the atmospheric pressure is less than the pressure of the system.

5 Question 5 10 / 10

✓ - 0 pts Correct

- 5 pts Unit of B is wrong.
- 0 pts You lost a "f" on A
- 5 pts Unit of B is wrong. $[B]=[P]^*[V]$
- 2.5 pts So what's the unit of B?
- 10 pts No answer submitted.

$$⑤ \quad p = A + (B/V)$$

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6 Question 6 10 / 10

✓ - 0 pts Correct

- 5 pts Wrong judgement on vacuum and gage pressure.
- 5 pts Wrong reading of Bourdon gage.
- 10 pts No answer submitted.

7

$$P_{atm} = 101 \text{ kPa}$$

$$v = 0.00122 \frac{\text{m}^3}{\text{kg}}$$

$$L = 12 \text{ cm} = 0.12 \text{ m}$$

$$\text{density } (\rho) = \frac{1}{0.00122 \frac{\text{m}^3}{\text{kg}}} = 819.7 \frac{\text{kg}}{\text{m}^3}$$

$$P_A = P_{atm} + \rho g(L+x)$$

$$P_B = P_{atm} + \rho g(x)$$

$$P_A - P_B = P_{atm} + \rho g(L+x) - P_{atm} - \rho g(x)$$

$$= \rho g(L+x-x) = \rho gL$$

$$P_A - P_B = 819.7 \frac{\text{kg}}{\text{m}^3} (9.81 \frac{\text{m}}{\text{s}^2}) (0.12 \text{ m}) = 964.9$$

yes

The pressure does decrease as the kerosene flows from a to b as the pipe diameter decreases.

Question 8) Korotkoff Sounds

The sphygmomanometer used to measure blood pressure. A cuff is placed around the arm and inflated until its pressure is over the systolic pressure. Then the air is released from the cuff. When the cuffs pressure matches the arterial systolic pressure, blood begins to flow. The snapping sounds heard while slowly letting the pressure out of the cuff are called "Korotkoff" sounds. (practicalclinicalskills.com) These sounds are used to measure two types of pressure.

The blood pressure reading has a systolic number and a diastolic number and is stated as systolic over diastolic. Systolic refers to the phase when the heart pumps blood, called the systole. Diastolic refers to the diastole which is the phase in between heartbeats, when the heart is at rest while it refills with blood (Shiel).

As the air cuff is released of pressure, the first sound heard is the systolic pressure. Soon, the sound dies down, which marks the diastolic pressure. (Shiel)

Korotkoff sounds are still being researched today. The underlying physiology is not yet completely understood. The main beliefs for the causes of these sounds which

7 Question 7 8 / 10

✓ - **2 pts** Wrong units of final results.

- **0 pts** correct

- **3 pts** Wrong judgement of pressure increase or decrease.

- **3 pts** Did not answer the second question, increase or decrease of pressure.

- **7 pts** Did not answer the pressure difference between a and b.

- **10 pts** No answer.

7

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$$P_A = P_{atm} + \rho g(L+x)$$

$$P_B = P_{atm} + \rho g(x)$$

$$P_A - P_B = P_{atm} + \rho g(L+x) - P_{atm} - \rho g(x)$$

$$= \rho g(L+x-x) = \rho gL$$

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As the air cuff is released of pressure, the first sound heard is the systolic pressure. Soon, the sound dies down, which marks the diastolic pressure. (Shiel)

Korotkoff sounds are still being researched today. The underlying physiology is not yet completely understood. The main beliefs for the causes of these sounds which

are guiding the research are that the sounds are created from fluid turbulence or possibly as a result of arterial wall oscillations. (Campbell)

Dr. Nikolai Korotkoff discovered these sounds, and he had the idea that the sounds are caused by the opening and closing of the blood vessel. The fact that there are multiple phases of the sounds can be accounted for by the “net transmural pressure” oscillating from positive to negative (Campbell). Transmural pressure is the difference in pressure between two sides of a wall or separator.

Proper technique is very important for taking blood pressure. It was found that different stethoscope positioning leads to mishearing the timing of the sounds, which leads to incorrect readings. It is widely accepted that the first sound is the systolic pressure, however the measurement of the diastolic pressure is debated. The argument is between whether the measurement should be taken when the sound muffles or when the sound disappears completely.

People commonly have their blood pressure taken using machines. These machines mostly rely on oscillometric methods, and these machines have been found to give overestimation of hypotensive blood pressures and underestimation of hypertensive blood pressures.

Sources:

Campbell, Miles. <https://www.ncbi.nlm.nih.gov/books/NBK539778/>

Practicalclinicalskills.com <https://www.practicalclinicalskills.com/sphygmomanometer>

Shiel, William C. Jr. <https://www.medicinenet.com/script/main/art.asp?articlekey=15771>

8 Question 8 18 / 20

- **0 pts** Well explained.
- **4 pts** Investigation on the physical basis of Korotkoff sound could be more detailed and specific.
- **4 pts** Investigation on the role of systolic and diastolic pressure on medical practice could be more detailed and specific.
- **2 pts** Investigation on the physical basis of Korotkoff sound could be more detailed and specific.
- ✓ - **2 pts** Investigation on the role of systolic and diastolic pressure on medical practice could be more detailed and specific.
- **2 pts** No references listed.
- **20 pts** No answer.
- **8 pts** No answer of the significance of these pressures on medical practice.
- **10 pts** The handwriting is hard to read. Also, the investigation could be more detailed and specific.